SE 216 – SOFTWARE PROJECT MANAGEMENT
SOFTWARE TOOLS DOCUMENT

PROJECT NAME: StorySphere: A Safe and Interactive Digital Platform for Writers and Readers
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TASK #	PROJECT TASKS WHICH REQUIRE SOFTWARE TOOL SUPPORT
1	Backend & Real-time System
2	AI Integration
3	Real-time Communication
4	OCR & Image Processing

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SOFTWARE TOOLS FOR TASK 1: Backend & Real-time System

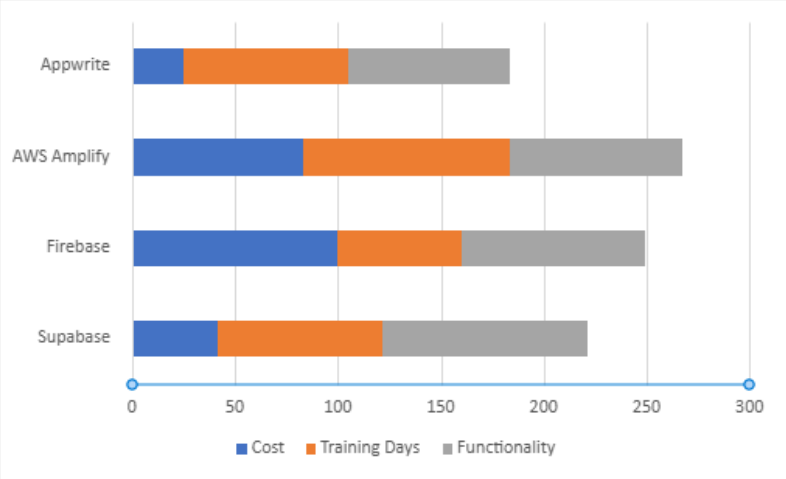
Tool Cost/Training/Functionality Data

Tool	Supabase	Firebase	AWS Amplify	Appwrite
Cost(month)	$25	$60	$50	$15
Training Days	4	3	2	4
Functionality	95	85	80	75

Normalized Cost/Training/Functionality Data

Tool	Supabase	Firebase	AWS Amplify	Appwrite
Cost	41,6	100.0	83.3	25
Training Days	80	60	100	80
Functionality	100.0	89.4	84.2	78.9

Normalized Tool Graph



Which tool has been selected? Why?

"Supabase has been selected for Task 1. When evaluating alternatives, Firebase's read/write-based model leads to unpredictable cost spikes (100.0). Supabase, on the other hand, protects our budget with its relational database (PostgreSQL) support and predictable pricing model (41.6). Thanks to its low vendor lock-in and the flexibility it offers for managing complex database relationships such as co-authorship and user systems, it provides the highest level of functionality (100.0) to our project's technical infrastructure."

SOFTWARE TOOLS FOR TASK 2: AI Integration

Tool Cost/Training/Functionality Data

Tool	OpenAI GPT	Google Translate	Azure AI	Anthropic Claude
Cost (month)	$20	$10	$30	$20
Training Days	2	1	4	2
Functionality	95	50	90	90

Normalized Cost/Training/Functionality Data

Tool	OpenAI GPT	Google Translate	Azure AI	Anthropic Claude
Cost	66.6	33.3	100	66.6
Training Days	50	25	100	50
Functionality	100.0	52.6	94.7	94.7

Normalized Tool Graph

Which tool has been selected? Why?
"OpenAI GPT has been selected for Task 2. Since Google Translate can only perform translations, its functionality score (52.6) remains low. While Azure AI is a strong alternative, its complex structure increases training time and costs (100.0). OpenAI GPT, on the other hand, reduces system complexity by offering all our AI features—such as writing assistance, translation, and content moderation—through a single API, and provides the highest level of natural language processing capability (100.0) for our project."

SOFTWARE TOOLS FOR TASK 3: Real-time Communication

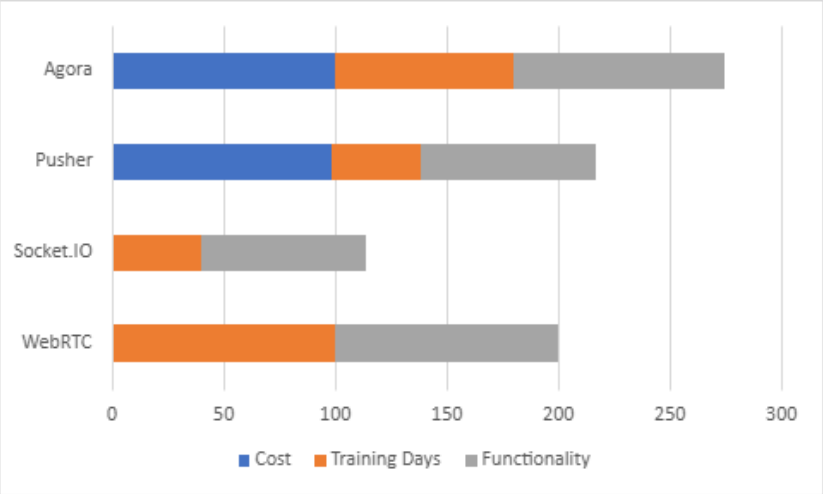
Tool Cost/Training/Functionality Data

Tool	WebRTC	Socket.IO	Pusher	Agora
Cost (month)	$0	$0	$49	$50
Training Days	5	2	2	4
Functionality	95	70	75	90

Normalized Cost/Training/Functionality Data

Tool	WebRTC	Socket.IO	Pusher	Agora
Cost	0	0	98	100
Training Days	100	40	40	80.0
Functionality	100	73.6	78.9	94.7

Normalized Tool Graph



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Which tool has been selected? Why?

"WebRTC has been selected for Task 3. Although Socket.IO and Pusher are quick-to-learn (40.0) tools for text-based communication, their functionality is insufficient for StorySphere's live streaming and co-session (audio/video) features. While Agora meets these requirements, it comes at a high cost (100.0). WebRTC, which is completely free and open-source (0.0), minimizes latency with its peer-to-peer architecture, providing our project with full functionality (100.0) in audio and video communication, despite having a longer learning curve (100.0)."

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SOFTWARE TOOLS FOR TASK 4: OCR & Image Processing

Tool Cost/Training/Functionality Data

Tool	Google Cloud Vision	Tesseract OCR	Amazon Textract	Azure Computer Vision
Cost (month)	$30	$0	$35	$25
Training Days	2	5	3	3
Functionality	95	50	85	90

Normalized Cost/Training/Functionality Data

Tool	Google Cloud Vision	Tesseract OCR	Amazon Textract	Azure Computer Vision
Cost	85.7	0	100	71.4
Training Days	40	100	60	60
Functionality	100.0	52.6	89.4	94.7

Normalized Tool Graph

Which tool has been selected? Why?

"Google Cloud Vision API has been selected for Task 4. Although traditional tools like Tesseract OCR are free (0.0), they fall short (52.6) in recognizing complex and irregular text structures like manga panels and require custom training (100.0). Amazon Textract, on the other hand, is more expensive (100.0). Google Cloud Vision is the most functional (100.0) solution, providing the highest accuracy with stylized fonts and detecting the content of speech bubbles, allowing us to successfully create our translation layers."